SmartCart AI

Frontend Behavioral Intelligence for Checkout Optimisation

1. **React**
2. **Tailwind CSS**
3. **Browser APIs**
4. **localStorage**
5. **Simulated AI Layer**

HACKATHON SUBMISSION 2026

Why Checkout Fails

70%

of online carts are abandoned before purchase — not from disinterest, but from friction, confusion, and lack of contextual guidance at the critical moment.

> "A user abandoning a ₹4,499 cart represents measurable, preventable revenue loss."

Hesitation During Checkout

Users pause, second-guess, and stall with no system response.

Coupon Hunting

Users leave to search for discounts and never return.

Shipping Confusion

Unexpected costs surface late, triggering drop-off.

No Contextual Guidance

Generic flows fail to adapt to individual user behaviour.

System Architecture

Fully frontend-driven, zero backend dependency — a deterministic event pipeline built for low-latency, browser-native execution.

- **User Layer**
- **Event Collection**
- **Hesitation Engine**
- **AI Rules**

Zero Backend

No server round-trips. All logic runs client-side.

Deterministic Pipeline

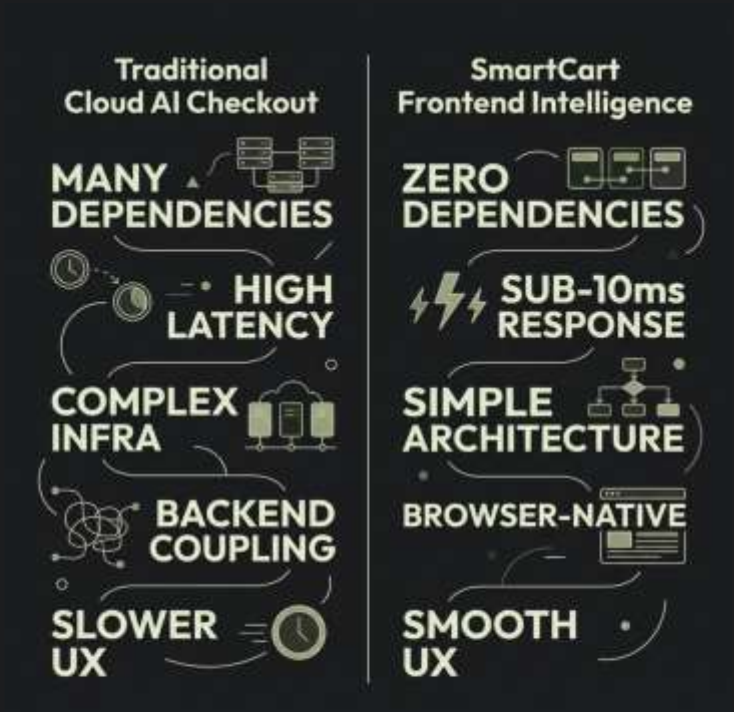
Predictable, auditable event-driven flow.

Browser-Native

Uses Web APIs for persistence and session continuity.

Frontend-First Intelligence

Choosing a frontend-first architecture was a deliberate engineering decision — not a constraint, but a strategic advantage for the hackathon context and beyond.



Rapid Development

No API contracts or infra provisioning to unblock progress.

Faster Interactions

Sub-10ms response — no network round-trip overhead.

Reliable Demo

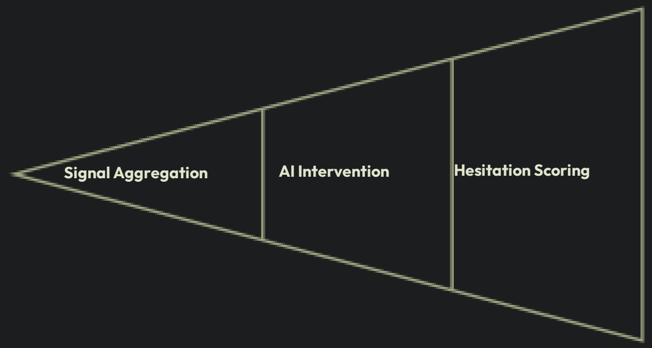
No live backend means no outages during judging.

Simpler Debugging

All state visible in browser DevTools — full transparency.

The Hesitation Detection Engine

A weighted scoring system monitors real-time user signals. Once the score breaches the threshold, targeted interventions fire automatically.



Idle ≥ 8s

+10 pts — passive disengagement signal

Tab Switch

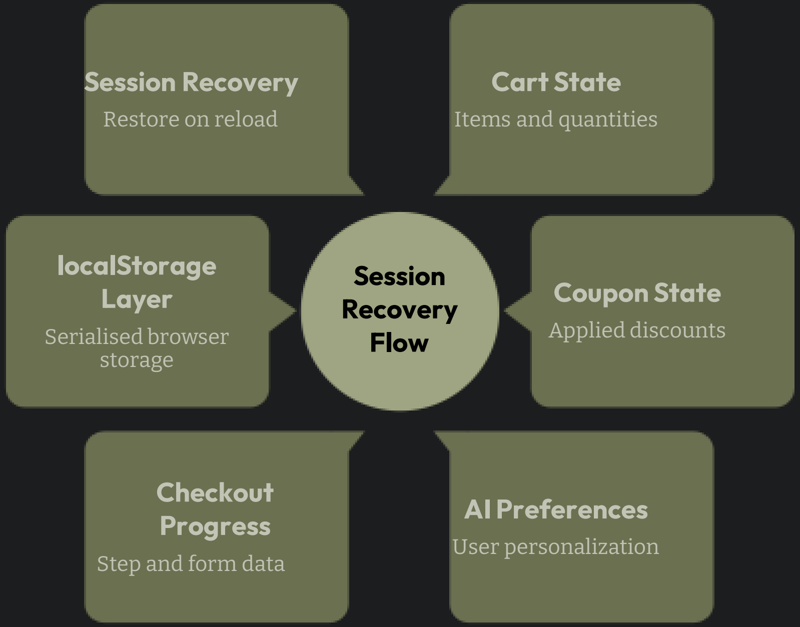
+25 pts — high-intent abandonment risk

Repeated Cart Edits

+15 pts — indecision pattern detected

Score ≥ 40 triggers: smart coupon reveal, urgency nudge, AI reassurance, or ₹600 savings insight.

State Management & Persistence



Browser-Native Continuity

All critical state is serialised to `localStorage` on every meaningful update — ensuring users never lose progress across refreshes or accidental tab closures.

Refresh Persistence

Cart and coupon state survive hard reloads.

Tab Recovery

Re-hydration on mount restores full session context.

Zero-Cost Storage

No database, no auth — lightweight and portable.

Performance & System Resilience

Rendering Performance

GPU-Friendly Animations

transform & opacity transitions only — no layout thrashing.

Minimised Re-renders

Memoised selectors and stable component boundaries.

Mobile Responsive

Tailwind breakpoints ensure smooth cross-device UX.

Failure Handling

Invalid Coupon

Silent fallback — user sees a helpful error, not a crash.

localStorage Unavailable

In-memory state takes over; no data loss visible to user.

Empty Cart Recovery

Redirect to product catalogue with preserved preferences.

No Recommendations

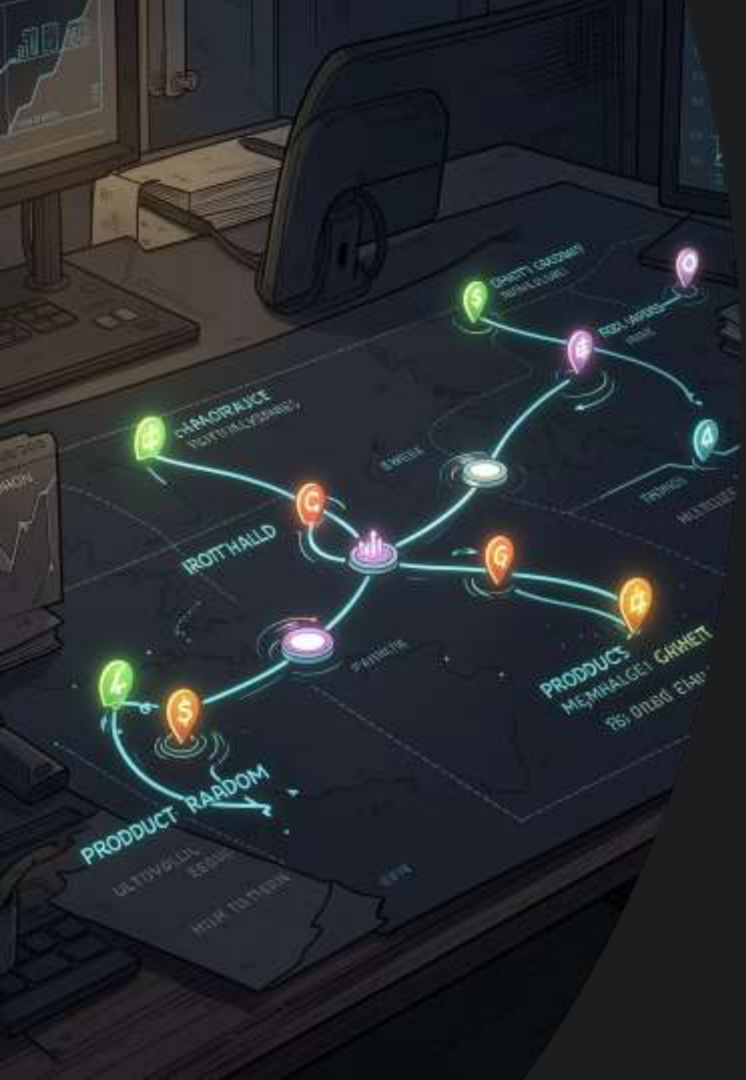
Default suggestion set renders instead of blank UI.

Design principle: **Graceful degradation over hard failure.**

Engineering Tradeoffs

Every architectural choice was made deliberately — optimising for demo reliability, build speed, and technical credibility within hackathon constraints.

Decision	Chosen Approach	Rationale	Tradeoff Acknowledged
ML vs Rules Engine	Deterministic rules	Auditable, zero training data required, instant iteration	Not adaptive; static scoring weights
Backend vs Frontend	Frontend-only	No infra overhead; reliable during live demo	Cannot persist cross-device
Database vs localStorage	localStorage	Zero-dependency persistence; browser-native API	5MB limit; no server sync
Live APIs vs Simulated	Simulated data layer	Eliminates third-party failure risk during judging	Not production-connected
Cloud AI vs Client-side	Client-side intelligence	Sub-10ms latency; works fully offline	No global model improvement loop



Scaling Beyond the Hackathon

The frontend-first foundation is a launchpad — each layer can be progressively replaced with production-grade infrastructure without breaking the core UX contract.

1

Phase 1

Shopify plugin + real payment integration (Razorpay/Stripe)

2

Phase 2

Cloud inference layer — replace rules engine with lightweight ML model

3

Phase 3

Merchant analytics dashboard — hesitation heatmaps, conversion funnels

4

Phase 4

Recommendation learning loop + multilingual support (Hindi, Tamil, Bengali)

Checkout Should Not Be Static

> "SmartCart AI prioritises responsive behavioural intelligence over infrastructure-heavy complexity."

Deterministic Intelligence

Predictable, auditable behavioural rules — no black-box ML.

Low-Latency UX

All interventions fire in-browser with zero network dependency.

Infrastructure-Light

Deploy anywhere — no cloud account, no API keys required.

Conversion-Focused

Every system layer exists to reduce abandonment and recover revenue.

REACT | TAILWIND CSS | BROWSER APIS | LOCALSTORAGE | HACKATHON 2026